

Powering Scalable Innovation: Streamlining Solar Pump Lifecycle Management with Upchain and Autodesk ERP Integration

- A Case Study

Executive Summary

Ecozen, a pioneer in climate-smart technology, developed **Ecotron**—a solar pump controller designed to revolutionize irrigation for Indian farmers. As demand is surging and production scaled to over **300,000 units**, Ecozen has been facing increasing complexity in product design, collaboration, change management, and deployment (Ecozen, 2024).

To address these challenges, there is a need of implementation of Upchain Product Lifecycle Management (PLM) (acquired by Autodesk) and integrated it with Enterprise Resource Planning (ERP) to streamline collaboration and product lifecycle management. This move established a digital thread across design, QA, pre-sales, and manufacturing operations, reducing time-to-market, improving collaboration, and delivering measurable return on investment (ROI).

The Challenges: Scaling with Disconnected Tools

Ecozen's Ecotron controller combines advanced MPPT algorithms, IoT data telemetry, and remote monitoring features—all within a compact unit installed beneath solar panels. Despite its engineering sophistication, the backend development and support processes were heavily dependent on manual tools and siloed workflows.

Specific pain points included:

- **Fragmented documentation:** Engineers managed Bills of Materials (BOMs) in Excel, design concepts in PowerPoint, and shared critical files via emails and cloud folders.
- **Lack of version control:** Different teams often worked with outdated or conflicting versions of design and test documents, leading to rework and confusion.
- **Manual change processes:** Change requests were handled informally, without structured approval or tracking systems.
- **Slow file access:** Locating the correct version of a design or test report could take hours across multiple storage locations.

These limitations became unsustainable as Ecotron moved from pilot to mass deployment. The lack of a centralized product development system meant that valuable time and resources were being spent chasing files, confirming revisions, and manually syncing data between systems.

The Solution: Integrated PLM and ERP Architecture

To address these challenges, implementation of PLM, selecting it for its cloud-native architecture, ease of integration, and user-friendly interface (Barrios, 2022). The PLM platform was integrated with Autodesk ERP, allowing for a continuous flow of information from product design to manufacturing execution.

Key elements of the solution included:

1. **Unified document management:** BOMs, CAD files, design notes, QA protocols, and test results were centralized in a single source of truth.
2. **Automated change management:** All Engineering Change Requests (ECRs) and Engineering Change Orders (ECOs) were tracked through formal workflows, with automated notifications and audit trails.
3. **ERP integration:** Approved BOMs and product data flowed directly into SAP, eliminating manual re-entry and ensuring procurement aligned with the latest design versions.
4. **Cross-functional access:** Sales, QA, and service teams gained real-time visibility into product data, improving pre-sales accuracy and post-sales support.

The PLM rollout was phased across engineering, QA, and operations teams. Customized dashboards were built for each role, making the adoption intuitive and aligned with existing workflows.

Implementation PLM process across Product

The implementation, as shown in Fig.1, begins by standardizing change management, with structured Engineering Change Requests (ECRs) and Engineering Change Orders (ECOs) routed through defined workflows. This centralized approach enabled faster approvals and reduced rework. The pre-sales team gains the access to accurate, up-to-date BOMs and configuration options, improving quotation accuracy and reducing miscommunication. During the design phase, real-time CAD collaboration and version control improved agility and reduced design errors. Integrated with QA workflows, the PLM system allowed quality assurance teams to tie test results directly to specific product versions for traceability. After design finalization, product data flowed into procurement and production via ERP integration, ensuring correct part sourcing and build plans. Once deployed in the field, support teams accessed design and change history for diagnostics, warranty validation, and service recommendations. As products aged, PLM utilizes to manage product obsolescence, ensuring compliance, spare part availability, and smooth transitions to newer variants.

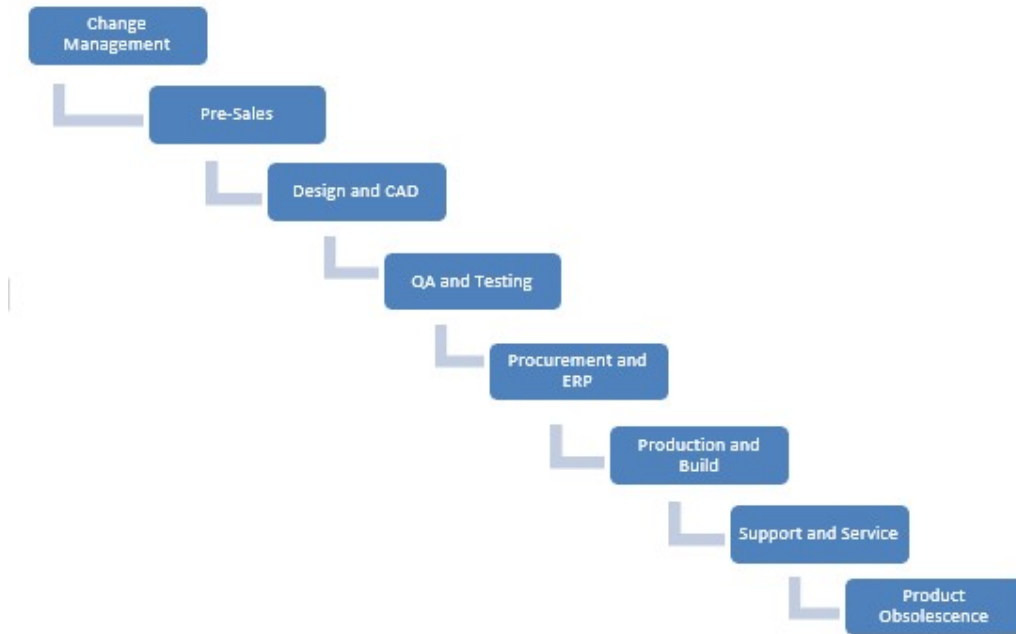


Fig.1 PLM-Driven Lifecycle

Results: Measurable Gains in Efficiency and ROI

The transition from manual tools to an integrated PLM–ERP ecosystem enabled Ecozen to work faster, smarter, and with fewer errors. Key performance improvements included reductions in rework, faster product launches, and greater throughput of new features and product variants.

The table 1 and table 2 highlight how PLM–ERP integration reduced delays, errors, and engineering rework, while accelerating approvals and product launches. This led to more efficient use of resources, increased throughput of new features and products, and faster time-to-market. Collectively, these improvements contributed to a measurable increase in operational efficiency and return on investment (ROI).

Table 1: Operational Efficiency

Metric	Before	After	Improvement
File retrieval time	2–3 hours	<10 minutes	~95% faster
BOM error rate	~15–20%	<2%	Significant accuracy gain
Design approval turnaround	5–7 days	<1 day	~85% faster
Product release cycle	6 months	4 months	33% faster

Table 2: Engineering and Innovation KPIs

KPI	Before PLM	After PLM	% Change
Engineering Rework	~30% of engineering time spent fixing issues	~10%	66% reduction

New Features Released / Quarter	4	6	50% increase
New Product Introductions / Year	2	3	50% increase

The PLM platform helped reduce repeated engineering work by centralizing version control, automating change approvals, and improving traceability. This allowed engineering teams to shift their focus toward innovation—leading to more features delivered and faster rollout of new Ecotron variants.

Improved collaboration across sales, QA, and design meant Ecozen could respond more quickly to field feedback, evolving customer needs, and system-level insights. Enhanced audit trails and change history also supported compliance and internal quality standards, essential in a fast-growing clean-tech sector.

Conclusion

Implementation of Upchain PLM—integrated with its ERP system—marked a strategic shift in how the company manages the lifecycle of its Ecotron solar pump. Transitioning from scattered tools like Excel and PowerPoint to a centralized digital thread enabled seamless collaboration across engineering, QA, pre-sales, procurement, and support teams. As a result, it will significantly reduce engineering rework, improved accuracy in design and BOM management, and accelerated both product releases and feature deployment. Although the performance metrics are based on internal estimates, they clearly demonstrate how PLM adoption unlocked measurable efficiency and innovation gains. This integration will position product to scale faster, respond more effectively to market demands, and bring greater agility to its clean-tech product development.

Key Takeaways

- **PLM–ERP integration** reduced manual errors, improved file traceability, and eliminated process redundancies.
- **Engineering efficiency** improved with up to 66% less rework and faster design approvals.
- **Innovation throughput** increased, with a 50% rise in new features and products.
- **Cross-functional collaboration** accelerated, enabling faster decision-making and customer responsiveness.
- **Digitized product lifecycle** management enhanced quality, compliance, and future scalability.

References

1. Ecotron: solar pump controller. (2024). <https://www.ecozensolutions.com/ecotron>

2. Barrios, P., Danjou, C., & Eynard, B. (2022). Literature review and methodological framework for integration of IoT and PLM in manufacturing industry. *Computers in Industry*, 140, 103688. <https://doi.org/10.1016/j.compind.2022.103688>